

Prasamita Bangal

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EDUCATION

Mahindra University	June 2028
<i>Integrated MTech in Computer Science</i>	<i>GPA: 8.0</i>
Delhi Public School (High School and Intermediate)	June 2023
	<i>GPA: 9.7</i>

COURSEWORK: Object-Oriented Programming, Data Structures & Algorithms, Database Management Systems, Discrete Math, Linear Algebra, GenAI and LLM, Artificial Intelligence and Machine Learning, Physics, Big Data Analytics, Recommender Systems, Wireless Sensor Networks

SKILLS

Programming Languages : Python , C , Java

Database : SQL , DBMS

Frontend : HTML , CSS , JS , React

Backend : Node.js , Flask

Cyber Security: Wireshark, Network Basics, Intrusion Detection , Drone Forensics, Real time anomaly detection, Security Protocols

Libraries : Numpy , Pandas , Tensorflow , Pytorch

Tools : Git/Github , Streamlit , Unix , Wireshark , VS Code , Scikit-learn , MATLAB , Network Basics

PROJECTS

- | | |
|---|------------|
| Drone Talk <i>Python, NLP, Streamlit, NumPy, Flask, Pandas, Git, VSCode, Unix</i> | Nov. 2025 |
| <ul style="list-style-type: none"> Built an NLP based drone command interface using Python and NLP libraries to convert natural language commands into structured drone control actions like “Fly 50 meters north and take a picture” into structured drone actions Implements intent classification, slot filling, and command generation for mission control | |
| Smart Medicine Return and Pharmacy Finder Platform <i>Flutter, Firebase, APIs, Node.js, React</i> | Sept 2025 |
| <ul style="list-style-type: none"> Developed a cross-platform application integrating Flutter frontend and Firestore backend for medicine return and pharmacy discovery. Implemented API integration and database querying concepts (DBMS, SQL) to fetch pharmacy data and manage user requests. | |
| AI-Driven Intrusion Detection System <i>Python, Matlab, Cybersecurity Basics, Data Analysis, SVM</i> | June. 2025 |
| <ul style="list-style-type: none"> Developed an AI-powered network intrusion detection system to detect and classify cyber threats in real time. Trained models using NSL-KDD dataset and achieved improved detection performance through optimized feature selection. Published at Jadavpur University Symposium (2024) following peer review | |
| Hybrid Predictive Maintenance using Enhanced NASA Dataset <i>Python, Pandas, Matplotlib</i> | June. 2025 |
| <ul style="list-style-type: none"> Designed an ML model for predictive maintenance of industrial equipment using NASA’s CMAPSS dataset. Focused on early fault detection using hybrid ML models. | |

EXPERIENCE

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| DLRL - DRDO <i>Project Trainee (Python, Scikit-learn, Matplotlib)</i> | Jan. 2026 – Present |
| <ul style="list-style-type: none"> Working on anomaly detection algorithms for UAVs. | |
| CSIR - IICT <i>Intern (PyTorch, NumPy, Data Analysis)</i> | June 2025 – Aug 2025 |
| <ul style="list-style-type: none"> Worked on analyzing real-time dengue patient data to identify and interpret emerging trends. | |

RESEARCH PUBLICATION

Drone Forensics Research | *Python , UAV Systems , Cybersecurity*

Ongoing

- Currently conducting research on drone forensics focusing on anomaly detection and malicious interaction in UAV systems

AI-Based Intrusion Detection System

2024

- Published a research paper at Jadavpur University Symposium on AI-Based Intrusion Detection System using machine learning techniques for network anomaly detection and cybersecurity analysis

EXTRA CURRICULAR

Student Placement Coordinator

- Assisted in coordinating placement-related activities and communication between students and placement cell

Student Council, Mahindra University | *Member*

- Sports Representative
- Coordinated sports-related activities and represented student interests in university sports initiatives
- Assisted in organizing campus sports events and coordinating in Inter University competitions
- **Skills gained:** leadership, coordination, and communication

Sports

- Chess
- Badminton
- Swimming

Social Service Initiatives

- Participated in Blood Donation Drive at Mahindra University
- Participated in Socio-Rural Development Event at Pune organized by Outreach Club, Mahindra University
- Active volunteer of Outreach Club, Mahindra University

AWARDS

- Scholarship for consecutive two academic years for good academic performance
- Top 10 in AI Student Of the Year Hackathon , Mahindra University
- Finalist in QubitZ Hackathon (Mahindra University , AEON)

HOBBIES

- Drawing
- Dancing (classical)
- Travelling to different places and driving